"MAJOR PROJECT TITLE"

Major Project Report

Submitted in Partial Fulfillment of the Requirements for the Degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

By

Name of Student
(Roll No.)

Under the Guidance of
Prof. _____________

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May 2026

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Acknowledgement

(Below is an exemplary Acknowledgement)

A journey is easier when you travel together. Interdependence is certainly more valuable than independence. This thesis is the result of work whereby I have been accompanied and supported by many people. It is a pleasant aspect that I have now the opportunity to express my gratitude for all of them.

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The chain of my gratitude would be definitely incomplete if I would forget to thank the first cause of this chain, using Aristotle's words, The Prime Mover for showering His blessings on me always.

(Name of the Student)

Abstract

(Below is an exemplary Abstract)

The explosive growth of wireless networks, ranging from Wireless Local Area Networks (WLANs) and Wireless Wide Area Networks (WWANs) to Mobile Ad hoc Networks (MANETs) indicates the beginning of "the wireless revolution". The proliferation of mobile computing and communication devices like cell phones, handheld digital devices, personal digital assistants, laptops or wearable computers is driving a revolutionary change in our information society. We are moving from the Personal Computer age (i.e., a single computing device per person) to the Ubiquitous Computing age in which a user simultaneously utilizes several electronic platforms through which he can access all the required information whenever and wherever needed.

Mobile Ad hoc Networks (MANETs) have become increasingly important as they provide ubiquitous connectivity which is not available with traditional fixed infrastructure networks. Such networks, consisting of potentially highly mobile nodes, have provided new challenges due to the unique characteristics of the wireless medium and the dynamic nature of the network topology. They require robust, adaptive communication protocols that can handle the unique challenges of these multihop networks smoothly. The TCP has been widely deployed as transport layer protocol on a multitude of internet works including the Internet, for providing reliable end-to-end data delivery. It is naturally viewed as the *de facto* reliable transport protocol for use in MANETs. As in the Internet it is desirable that TCP must provide reliable data transfer services for communication within wireless networks and also between wireless networks and wired Internet. However, assumptions made during TCP development reflected characteristics of the prevalent wired infrastructure of networks at the time and subsequently leads to sub-optimal performance when used in wireless ad hoc environments as observed in the simulation results. Transport protocol in an ad hoc network must handle mobility-induced disconnection and reconnection, route change-induced packet out-of-order delivery for mobile hosts, and error/contention-prone wireless transmissions. Modifying TCP to improve its performance in wireless networks has been a long-standing research problem. Many methods have been proposed to improve TCP's performance in MANETs. Among them Ad hoc TCP (ADTCP) uses an end-to-end approach which is easy to implement and deploy since it requires minimal changes at the sender and receiver, provides the flexibility for backward compatibility and is TCP-friendly. To further improve the performance of ADTCP we consider following: ensure sufficient bandwidth utilization of the sender-receiver path, avoid the overloading of network and

check for incipient congestion and take appropriate actions. For this we limit TCP's congestion widow below the BDP-UB (Upper Bound of BDP) of sender-receiver path, i.e its maximum packet carrying capacity.

The project is an attempt to improve the performance of ADTCP in MANETs by estimating the optimum window size and then setting congestion window limit to an optimum value.

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LIST OF FIGURES

LIST OF TABLES

NOMENCLATURE

A	Energy level indicator
B	Bottom product rate, kmol/hr

Greek

θ	Root of Underwood equation
α	Relative volatility
λ	Latent heat of vaporization, kcal/kmol
Δ	Difference
ε	Energy change indicator

Subscripts

min	Minimum
i	Any component

Abbreviations

CGCC	Column Grand Composite Curves
IRS	Invariant Rectifying Stripping
CMO	Constant Molar Overflow

Chapter 1

Introduction

1.1 Prologue

MANETs have become increasingly important in view of their promise of ubiquitous connectivity beyond traditional fixed infrastructure networks. They represent complex distributed systems that comprise wireless mobile nodes that can freely and dynamically self-organize into arbitrary and temporary, ‘‘ad hoc’’ network topologies **[1].**

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Examples:

[1] B. Klaus and P. Horn, *Robot Vision.* Cambridge, MA: MIT Press, 1986.

[2] L. Stein, "Random patterns," in *Computers and You,* J. S. Brake, Ed. New York: Wiley, 1994, pp. 55-70.

[3] R. L. Myer, "Parametric oscillators and nonlinear materials," in *Nonlinear Optics*, vol. 4, P. G. Harper and B. S. Wherret, Eds. San Francisco, CA: Academic, 1977, pp. 47-160.

[4] M. Abramowitz and I. A. Stegun, Eds., *Handbook of Mathematical Functions* (Applied Mathematics Series 55). Washington, DC: NBS, 1964, pp. 32-33.

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[1] *Name of Manual/Handbook*, *x* ed., Abbrev. Name of Co., City of Co., Abbrev. State, year, pp. *xx-xx.*

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[1] *Transmission Systems for Communications*, 3rd ed., Western Electric Co., Winston Salem, NC, 1985, pp. 44–60.

[2] *Motorola Semiconductor Data Manual*, Motorola Semiconductor Products Inc., Phoenix, AZ, 1989.

[3] *RCA Receiving Tube Manual*, Radio Corp. of America, Electronic Components and Devices, Harrison, NJ, Tech.Ser. RC-23, 1992.

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[2] J. H. Davis and J. R. Cogdell, "Calibration program for the 16-foot antenna," Elect. Eng. Res. Lab., Univ. Texas, Austin, Tech. Memo. NGL-006-69-3, Nov. 15, 1987.

[3] R. E. Haskell and C. T. Case, "Transient signal propagation in lossless isotropic plasmas," USAF Cambridge Res. Labs., Cambridge, MA, Rep. ARCRL-66-234 (II), 1994, vol. 2.

[4] M. A. Brusberg and E. N. Clark, "Installation, operation, and data evaluation of an oblique-incidence ionosphere sounder system," in "Radio Propagation Characteristics of the Washington-Honolulu Path," Stanford Res. Inst., Stanford, CA, Contract NOBSR-87615, Final Rep., Feb. 1995, vol. 1.

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[2] N. Kawasaki, “Parametric study of thermal and chemical nonequilibrium nozzle flow,” M.S. thesis, Dept. Electron. Eng., Osaka Univ., Osaka, Japan, 1993.

[3] N. M. Amer, “The effects of homogeneous magnetic fields on developments of tribolium confusum,” Ph.D. dissertation, Radiation Lab., Univ. California, Berkeley, Tech. Rep. 16854, 1995.

[4] C. Becle, These de doctoral d’etat, Univ. Grenoble, Grenoble, France, 1968.

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Appendix

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